

An Educational Game Prototype for Avatar-based Non-Manual Sign Learning in American Sign Language

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Abstract

American Sign Language (ASL) learners often struggle with mastering non-manual signs (NMS), such as facial expressions, which are essential for ASL grammar. This study introduces CoSignPlay, an avatar-based ASL learning game designed to emphasize NMS through a collaborative approach, where learners work in pairs to separately learn manual signs (MS) and NMS. We conducted a design probe study with two ASL instructors to explore common NMS teaching practices and gather feedback on key design elements, including the game, 3D avatars, and collaboration. Results showed that participants recognized the importance of NMS but expressed concerns about the challenges of learning and teaching them. They acknowledged the potential of avatars and collaborative learning to clarify NMS but raised concerns about avatars' naturalness and the effectiveness of separating sign components into MS and NMS. This work contributes to understanding how avatar-based learning games and collaborative strategies can enhance ASL education, particularly in NMS.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

American Sign Language, non-manual signs, avatar, collaborative learning, deaf and hard of hearing

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1 Introduction

One common misconception about sign language is that it is the “language on the hands”. Non-manual signs (NMS), including facial expressions, head tilts, mouthing, and other non-manual features, are equally important as manual signs (MS) [37]. They are the essential components of American Sign Language (ASL) grammatical expressions to create or influence meaning [16, 37]. For example, in ASL, the signs for “HERE” and “NOT HERE” use the same MS but are distinguished only by NMS [3]. Moreover, previous research indicates that both ASL instructors and learners identify NMS as one of the most challenging aspects of learning ASL, with difficulties arising from cultural inhibitions, physical coordination, and a lack of understanding of its grammatical role [26]. Consequently, to facilitate a comprehensive understanding of signs and support ASL learners at all proficiency levels, it is important that future ASL educational tools emphasize NMS as an integral component of ASL instruction.

Prior work on learning technologies that facilitate ASL learning mainly focuses on MS [7, 10, 30, 36], with NMS largely neglected. There is a body of game-based ASL learning technologies [6, 10, 36]. For example, Weaver et al.'s [36] study found that children obtain better ASL language acquisition using educational computer games compared to normal classroom curriculum. Developing game-based learning where ‘learning through the game’ also resonates with the experiential learning theory, which highlights the importance of engaging learners in hands-on experiences and reflection [18]. Moreover, there is emerging research on using 3D avatars to support ASL learning [30, 38]. Studies indicate that avatar-based ASL learning not only enhances realism, engagement, and comprehension across diverse groups compared to traditional sign notation [2, 9, 15], but also improves digital access, supports sign language learning, and facilitates signed communication [28, 31]. Therefore, we are motivated to investigate the unique needs and challenges of ASL NMS learning, and explore the design opportunities of game and avatar-based learning technologies in providing efficient learning experiences for ASL learners.

Collaborative learning is an educational approach where two or more individuals work together to learn or attempt to learn something by engaging in interaction and leveraging each other's knowledge and skills [4]. Previous studies have shown that this learning strategy can enhance social interaction among Deaf and Hard-of-Hearing (DHH) children, support shared knowledge construction, and improve the quality of knowledge [4, 14]. The aforementioned game and avatar-based ASL learning technology studies [7, 10, 30, 36] primarily focus on individual use by deaf children, overlooking the potential benefits of shared responsibility and dual-input learning experiences that a collaborative learning approach could offer. To address this gap, we propose co-signing activities in which pairs of learners focus on separate aspects—MS and NMS—while engaging with each other in communication, promoting both individual learning and collaborative interaction.

To better understand the design space for learning technologies that support NMS learning, we aimed to interview ASL instructors who have a deeper understanding of how learners acquire ASL and answer the following research questions:

- How do ASL instructors perceive the role of non-manual signs (NMS) in ASL education?
- How do ASL instructors perceive the role of avatars in demonstrating NMS in ASL learning?
- How do ASL instructors perceive the effectiveness of collaborative learning, where MS and NMS are learned separately, in improving ASL learning?

To answer these questions, we conducted a design probe study to investigate common teaching practices of ASL NMS and feedback on key design elements that involve games, 3D avatars, and collaboration. Additionally, we explored the potential benefits of these design elements in supporting a special case of collaborative learning between hearing parents and their DHH children, both of whom may learn ASL simultaneously to enrich their everyday interactions and promote positive child development. We interviewed two teachers of ASL programs for adult learners in the USA. We found that participants highlighted the importance of emphasizing NMS in ASL, while also expressing concerns about the challenges associated with learning and teaching NMS. They showed positive attitudes toward avatar-based technology and the concept of collaborative learning games but raised concerns about the naturalness and clarity of the avatar's features, as well as the approach of teaching MS and NMS components separately. Overall, our study provides valuable insights into the potential of avatar-based technologies to support collaborative learning in ASL education.

2 Related work

2.1 Non-manual signs and learning challenges

NMS is an important part of sign language. It includes facial expressions, head movements, and body positions that provide essential grammatical context to the MS [37]. NMS serves various grammatical functions, including lexical, morphological, and syntactic [29]. Taking syntactic NMS as an example, which is frequently used in ASL, they play a critical role in defining sentence types, including Yes/No questions, negation, and affirmation [17]. For instance, the signer needs to raise their eyebrows for "WILL YOU HELP ME?" and maintain a neutral face for "YOU HELP ME," even though the

MS remains the same [35]. Therefore, mastering NMS is essential for fully understanding the meaning that signers convey.

In natural signed communication, NMS are everywhere and integral to meaning. Research has demonstrated that during the communication between signers, they primarily focus on each other's faces instead of their hands, indicating the important role of NMS [34]. Unlike MS, which are easily visible through hand movements, identifying and analyzing NMS is more challenging due to their often subtle characteristics [29]. Additionally, unlike spoken communication where voices are sequential, NMS often occurs simultaneously with different hand shapes, orientations, and types of movement in different locations, making it challenging for people to decode all elements at the same time [27]. These challenges highlight the complexity of recognizing and learning NMS in ASL, motivating ASL educators or technology designers to pay special attention to the teaching of NMS.

2.2 Technology tools in ASL education

There are many technological tools for learning ASL, such as games and avatars [7, 10, 30, 36]. Aligning with the Experiential Learning Theory, which highlights the importance of engaging learners in hands-on experiences and reflection [18], games have been acknowledged as an effective tool for fostering these experiences in language learning and developmental processes [13, 18, 32]. The CopyCat project is a typical example of an ASL learning game where DHH children communicate with the computer using ASL. Specifically, they are tasked with solving a problem by telling the game's mascot what to do in ASL [7, 36]. If the children provide ASL description correctly, the game scenario advances to the next scene. Findings show that game-based learning enhances children's language acquisition compared to traditional classroom curriculum. However, in this game, if children perform incorrect signs, the game will block users from proceeding. This game design somewhat creates a sense of frustration or disengagement for learners if they are unable to progress. To improve this, incorporating alternative pathways might allow the learners to continue engaging with the game while still reinforcing the correct learning.

Signing avatars are another popular technological tool to help people learn ASL, offering natural language representation and ensuring accessible information for the signing deaf community [1, 9]. From learning aspects, such avatar-based technology could also increase learners' content comprehension, willingness to take an active role, and learning engagement [23, 25, 38]. Quandt et al. [30] used signing avatars created from motion capture recordings of DHH signers to teach ASL in an immersive VR environment and they received positive feedback from learners. However, this study seems to facilitate MS acquisition primarily and neglects the equally critical role of NMS. It was found that NMS in sign language avatars convey linguistic and emotional meaning and improve the realism of the avatars, beyond what MS alone can achieve [5, 22]. Therefore, future studies should ensure that the design space of the avatar in the ASL learning game gives equal attention to both MS and NMS, recognizing their critical importance.

2.3 Collaborative learning

Previous studies found that in collaborative learning, the success of one participant contributes to the success of others as individuals are responsible for their own learning as well as the learning of their peers [20]. A meta-analysis has also shown that collaborative game-based learning enhances both learning outcomes and motivation [39]. Therefore, this strategy may yield promising outcomes for ASL acquisition. To date, researchers have attempted to evaluate the impact of digital collaborative interaction. Bottoni et al. [8] developed the CoPuppet project, which allowed multiple performers to manipulate different parts of a virtual puppet through gestures and voice. This introduces a new collaborative educational tools design and innovation. In recent years, the development of interfaces has made it possible to integrate multiple individuals into a single avatar, enabling the combination of skills and the distribution of cognitive load in controlling virtual avatars [19, 21]. Therefore, co-signing activities on a virtual avatar might bring a promising learning outcome. Each user would be responsible for different components (i.e., MS and NMS) of signs. They learn and communicate by engaging with each other to manipulate the performance of the virtual avatar. This co-signing activity could not only be supported by the benefits of collaborative learning shown in previous studies [20, 24, 39], but also could be supported by the cognitive load theory which illustrated that processing new information requires mental effort resulting in cognitive load on working memory and then affecting learning outcomes [33].

3 Design and implementation of CoSignPlay

To address the research gaps in previous studies, we proposed CoSignPlay, an avatar-based ASL learning game, to improve both MS and NMS acquisition through collaborative gameplay. The game integrates different daily-life scenarios where players interact with game characters using ASL.

3.1 Game concept

In CoSignPlay, players are put into several daily-life scenarios to communicate with other in-game characters using ASL, such as interacting with a deaf restaurant waiter to order food (Fig.1(a)). Before requesting the players to sign an ASL sentence, the game introduces the basic ASL knowledge required, ensuring players understand the MS and NMS they will use. When players sign the required sentence correctly, the story advances (e.g., the customer enjoys the meal) (Fig.1(b)), and new challenges are introduced. However, when players fail to produce the sign accurately, the in-game character misunderstands what the avatar shows, instant feedback about the incorrect signs will appear, and then the game will proceed to the undesired outcome (e.g., the customer has an allergy reaction) (Fig.1(c)).

CoSignPlay includes four basic types of NMS [17, 29] - Yes/No question ("Is the oven on?"), negation ("I don't need tomatoes because of an allergy"), degree ("I'd like a large pizza, or a larger pizza"), and lexical ("I want vanilla ice cream").

3.2 Avatar control modes: single vs. two players

CoSignPlay introduces two avatar control modes: (1) single player - the MS and NMS from a single player will be mapped to the avatar

(Fig.2(a)); (2) two players - the MS produced by one player and NMS produced by another player are jointly mapped to the avatar (Fig.2(b)). The two players have to coordinate to jointly complete ASL sentences. This division of responsibilities encourages players to work closely together, fostering communication and mutual understanding as they navigate the game's challenges. For example, in a scenario where the game requires players to sign "I don't understand." The player handling MS must accurately sign "understand," while the NMS player must shake their head and make a questioning facial expression synchronously. Only by aligning their inputs seamlessly can the players successfully deliver the intended meaning of the sentence.

3.3 Implementation of avatar mapping

CoSignPlay integrates an avatar mapping system that enables players to interact with the game using intuitive, accessible technology. To accommodate a wide range of users, a standard webcam is used as the motion capture tool without any specialized equipment. While a dedicated motion capture system might provide better precision of motion tracking, the convenience and accessibility of the game are considered to be more important to make it available to a broader audience.

The system utilizes a machine-learning-based framework, MediaPipe, to detect and track players' body movements and facial expressions. Players' real-time videos are captured by the webcam and processed by MediaPipe to identify key body landmarks, such as joint positions and facial regions, and translate these coordinates into real-time animations of the in-game avatar. A series of scores derived from motion capture data is used to represent different facial movements. These scores correspond to specific features such as eyebrow raise, mouth shape, or head tilt and are applied to control the avatar's facial expressions, making them align with the captured human expressions.

4 Research method

We are in the process of conducting a semi-structured interview study with ASL instructors to investigate the teaching practices of NMS and feedback on the key design elements of CoSignPlay, including the game concept and the single-player and two-player avatar control modes.

4.1 Participants

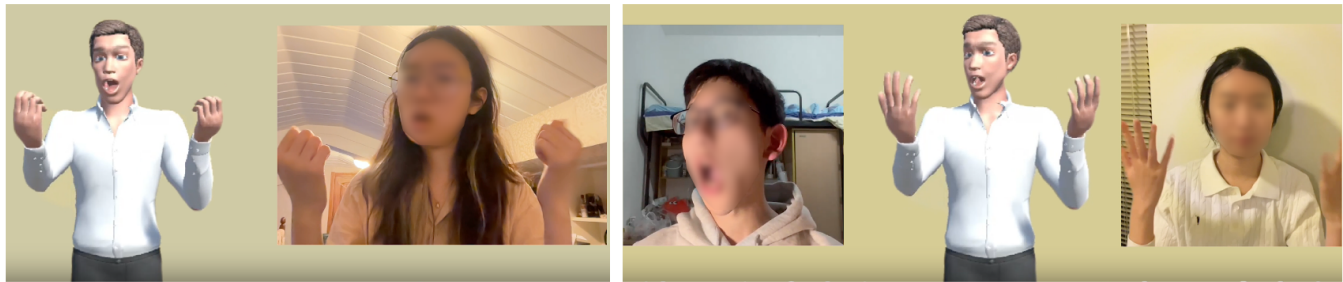
We conducted a pilot interview with one ASL instructor who is a member of the research team. Based on the positive feedback, we interviewed two deaf or hard-of-hearing ASL teachers (1 male and 1 female; mean age = 53.5 years) recruited through study flyers and snowball sampling. Both participants teach ASL programs for adult learners and have more than five years of teaching experience. They primarily use sign language as their daily communication method. The compensation for the study is \$30/hour for participating in a remote interview lasting between 1.5 to 2 hours. Both participants consented to the information sheet before the study and this study was approved by the Institutional Review Board of the researchers' university.

Figure 1: An illustration of the game concept of CoSignPlay

(a) The customer don't want tomato on the pizza.

(b) The avatar makes correct NMS. The customer gets the pizza without tomato and enjoys the meal.

(c) The avatar makes incorrect NMS. The customer gets the pizza with tomato and gets an allergy reaction.

Figure 2: An illustration of the avatar mapping.

(a) One player controls both MS and NMS.

(b) Player 1 controls NMS and Player 2 controls MS.

4.2 Study procedure

The interviews consisted of three sections and were conducted in a semi-structured way.

4.2.1 Section 1: background discussion. Participants reflected on their NMS teaching practices and strategies, shared perspectives on collaborative and playful methods for teaching NMS in the classroom, and evaluated the potential benefits and limitations of technology in ASL education.

4.2.2 Section 2: game design and avatar control feedback. After watching the demonstration videos of the game's single-player and collaborative modes, participants were asked to evaluate the core design mechanisms, including mirroring signs to an avatar and the concept that incorrect signs result in undesired outcomes. They also discussed the appropriateness of the chosen NMS categories, identified potential benefits and concerns related to the collaborative learning approach where one player controls NMS and the other controls MS, and suggested possible improvements or alternative game scenarios.

4.2.3 Section 3: prototype interaction. Participants engaged with the game prototype to experience its core mechanics, focusing on the avatar mapping system. This system allowed players to control a 3D avatar and observe how it tracked and represented their

movements in real time. The prototype aimed to introduce participants to the concept of avatar-based collaborative learning games specifically designed to support the acquisition of NMS in ASL. Through hands-on interaction, participants explored the potential of this approach to enhance learning and provided feedback on its educational value.

4.3 Data analysis

Interviews were video-recorded and transcribed anonymously by a third-party transcription service following each session. Transcription files were imported and managed using NVivo version 15. Inductive thematic analysis [11] was used to analyze the interview transcript. This method begins without predetermined frameworks, offering a 'starting point' for the flexible analysis of data and enabling themes to be constructed through concept synthesis [12]. Two researchers actively read through the transcripts and independently developed codes that related to the context of research questions (NMS learning, avatar-based technologies, and collaborative learning methods). To ensure the validity of the analysis, two researchers met and discussed the emerging codes and addressed discrepancies between codes. These codes were synthesized into themes based on conceptual similarities.

5 Results

5.1 Insights and challenges regarding NMS learning

All participants highlighted the importance of emphasizing NMS in teaching and learning ASL. P1 noted, *“If you’re teaching signs without the facial expressions and the other that go with it, they’re not learning everything.”* Specifically, *“I do believe that facial expressions and non-manual signs are 70% of the part of the [sign] language and grammar.”* (P2). For example, *“there’s different mouth shapes or movements that express size, emotion, movement”,* indicating that *“[NMS] can affect what a sign means.”* (P2)

However, participants expressed a few concerns about the challenges that learners face while learning NMS. One common difficulty is students *“overlook [facial expressions] or their head movements”* (P2) or *“forget in between what they’re learning and that applies to”* (P1). This unconscious omission might contribute to the disconnect between learning individual components of ASL and applying them effectively in conversation. Besides, all participants mentioned that students have diverse learning paces, and some students learned in slow and struggling ways due to cognitive processing. For example, P1 noted that *“[while] the cognitive processing [of some DHH children] allows them to pick [NMS] up quickly, there are other folks that struggle ... to get that fluidity [and] fluency.”* Plus, *“some people struggle with [combining head movements and eyebrow motions] at the same time.”* (P2).

In addition to the learner’s personal challenges, participants showed a few reservations about the effectiveness of teaching DHH children. P1 illustrated some NMS teaching materials they employed in the classroom are limited in terms of dimensionality and interaction, *“Learning from [ASL textbook] that is two-dimensional is not great ... [even though] watching [ASL] videos is a good way to learn but [there is no interaction].”* Furthermore, the lack of formal ASL education makes the learning process harder. P1 highlighted that some learners learned ASL *“from people that have maybe [incorrect] homemade signs”,* and this often leads to difficulties in learning as they have to *“unlearn those [incorrect signs] and learn the correct signs.”*

5.2 Benefits and challenges of avatar-based learning

After viewing the demo video of CoSignPlay and trying the avatar tracking system, participants expressed positive feelings toward the avatar-based learning approach. P1 highlighted the issue of *“people can’t see themselves [NMS and MS signing]”* could be potentially solved by *“[if learners] see the avatar doing what they’re doing, they’re quicker to recognize their own mistakes and correct it, and maybe they’ll learn faster.”* Additionally, the avatar provides a safe and comfortable environment for learners to practice by allowing them to compare their signing with the avatar’s demonstration without the pressure of *“seeing their own face [making mistakes]”*. Participants also praised the potential abilities of such avatar-based technology to *“reinforce what [learners] have learned [in the class]”* (P1) and take the role of *“supplemental teaching”* to *“help [learners] practice on their own”* in the *“privacy of their room.”* (P2)

However, they also showed a few concerns about avatar-based learning. Some of them worried that the avatar tracking of facial expressions and finger spelling needs to be very clear and natural to have the same effects as what a real person did. For instance, P2 illustrated that *“if I ... make a slight [mouth] movement, ... the technology can’t grasp it. ... [When] something’s large and makes this mouth morpheme, [if] the avatar just opens its mouth, it’s not the same. There’s a disconnect.”* Besides, while avatars can effectively convey certain ASL elements, participants still might not yet be preparable for avatar-based teaching or learning. For example, P1 emphasized, *“it is still new to the students ... [they] prefer a real live person in front of them signing”*. Moreover, P2 believed, *“[a people’s video recording] would be a better connection”* compared to avatar-based teaching.

5.3 Feedback on collaborative NMS and MS learning approaches

Participants generally conveyed positive sentiments about the collaboration concept. They conveyed that in the regular classroom, one effective teaching method is *“pairing [students] up”* (P2) and *“[students giving] feedback to each other and [doing] a dialogue.”* (P1) They felt that instant feedback from the learner’s partner during the collaboration activities could enhance their ASL learning. Additionally, the cognitive overload when learning NMS and MS simultaneously could be potentially solved by the collaborative method. *“You start with just the non-manual signs, and as you [switch roles with your partner], you add in the manual signs, so that’s not so overwhelming.”* (P1) In addition to cognitive overload, the separating learning method could allow learners to understand the effect of various NMS on the meaning of the same MS comprehensively. For example, they could collectively explore how NMS adds layers of meaning, as P2 stated, *“There may be only one or two meanings applied to a manual sign. So working on them separately, you can do more with the non-manual signs, you can switch it out for more and more and more meaning.”*

However, participants had several concerns regarding the collaborative game. They believed that separating NMS and MS during the learning process could negatively affect the fluidity with which learners apply ASL in daily life. As P1 noted, *“It is important that [NMS and MS] go together and that they are not separate so that it’s a little bit more fluent to understand.”* Moreover, participants emphasized that if in the case of hearing parents and DHH children, hearing parents might be more comfortable with lip-reading as a primary strategy for communication and less willing to use NMS during collaboration with their DHH children. *“Sometimes the [hearing] parents just want to talk and they don’t want to do the facial expressions. ... They want to have [DHH children] read lips.”* (P2)

6 Conclusion

In this study, we proposed CoSignPlay, an ASL learning game that focuses on NMS and integrates an avatar-based collaborative learning experience, to better understand the design space for learning technologies supporting NMS acquisition. The findings showed that participants expressed positive opinions toward using avatars

to help ASL learners quickly identify any mistakes in their signings and provide an environment where they can practice without the anxiety of making mistakes in real-life conversations. However, concerns about the accuracy of the motion capture technology were stated. The avatar should look natural and display both MS and NMS signals clearly, minimizing the risk of misunderstanding. As for the collaborative learning design, participants supported the idea of pairing learners together so they could give each other feedback during the collaboration. Separating MS and NMS during the learning process was also considered a way to ease the difficulties of learning MS and NMS simultaneously. While this approach is a novel idea, participants also showed concerns regarding whether learners can integrate MS and NMS in a real-life signing context after practicing them separately.

As this is a preliminary study exploring the concepts of NMS, avatar integration, and collaborative learning, we plan to recruit more participants to gain comprehensive insights into ASL NMS education. Future studies should focus on refining the game design further and supporting diverse ASL learning goals. Expanding the participant pool to include ASL learners, as well as hearing parents and DHH children, will be essential for validating and improving these approaches across different learning contexts.

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